

**People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research**



University of Science and Technology Houari Boumediene

**Faculty of Computer Science
Department of AI and Data Science**

Master's Degree in Intelligent Computer Systems

**Master's Thesis
Automatic Recognition of Qur'anic Tajweed
Recitation**

Prepared by:

Haddad Ahmed Ayoub — 212136035139

Kessal Haroune — 202033043645

Academic Year: 2025 / 2026

Contents

1	State of the Art	2
1.1	Introduction	2
1.2	Qur’anic Recitation, Tajweed, and Learning Support	4
1.2.1	Qur’anic Recitation and Oral Transmission	4
1.2.2	Qirā’āt, Riwayāt, and Recitation Variants	4
1.2.3	Fundamental Tajweed Rules	4
1.2.4	Types and Severity of Recitation Errors	6
1.2.5	Traditional Learning and Teacher Correction	6
1.2.6	The Coloured Mushaf as a Visual Learning Aid	7
1.3	Resources, Related Work, and Research Gap	9
1.3.1	Speech and Language Resources for Qur’anic Recitation	9
1.3.2	General Landscape of Related Work	10
1.3.3	Existing Approaches to Qur’anic Recitation Assessment	10
1.3.4	Evolution of Approaches	12
1.3.5	Comparative Summary of Reviewed Systems	13
1.3.6	Comparison of Model Architecture Options	14
1.3.7	Research Gap and Positioning of the Present Work	14
1.3.8	Chapter Summary	15
	Bibliography	18

Chapter 1

State of the Art

1.1 Introduction

The Qur'an occupies a central place in Islamic worship, learning, and daily practice. Its recitation is not treated as ordinary reading: it is a transmitted oral practice in which pronunciation, rhythm, pausing, elongation, and articulation must follow recognised rules. This discipline is known as Tajweed. For learners, Tajweed is often one of the most challenging aspects of Qur'anic study because errors are primarily acoustic: a mistake may be entirely invisible in the written word yet immediately audible in how a sound is produced.

This chapter establishes the background and context needed to understand automatic Tajweed assessment. It begins with an overview of Qur'anic recitation and the system of canonical transmission traditions known as *Qirā'āt* and *Riwayāt*. It then presents the main Tajweed rule families and their acoustic relevance, and discusses the types and severity of recitation errors, because not all mistakes carry the same weight or require the same form of analysis. The chapter also examines the pedagogical role of traditional teacher correction and of the coloured *Mushaf*, a widely used visual learning aid, as these two supports define what an automated system must eventually replace or complement.

The second part of the chapter turns to the state of the art. It surveys the main categories of resources available for automatic Tajweed assessment, reviews the principal research directions and their representative systems, and compares the modelling approaches found in the literature. The objective is not merely to describe existing work, but to identify the gap that this thesis addresses. As the review will show, current systems provide valuable but partial solutions: they perform speech recognition, classify specific rules, or detect phoneme-level errors. However, a complete automatic Tajweed assessment system must achieve more: it must verify the recited content, locate each deviation, identify the applicable rule, distinguish the type of mistake, and produce feedback that is actionable for the learner. No existing system achieves all of these simultaneously.

The chapter is organised as follows. Section 1.2 covers the Qur'anic recitation background,

Tajweed rule families, error types, and pedagogical learning supports. Section 1.3 presents the resources available for automatic assessment, reviews existing approaches by technical direction, compares model architectures, and identifies the research gap addressed by this thesis.

1.2 Qur’anic Recitation, Tajweed, and Learning Support

1.2.1 Qur’anic Recitation and Oral Transmission

The Qur’an is the sacred text of Islam and its recitation occupies a central place in Islamic worship, education, and oral tradition. Unlike ordinary reading, Qur’anic recitation is governed by a precise set of phonetic, rhythmic, and articulatory constraints. The learner is expected not only to pronounce the Arabic text correctly, but also to preserve the manner of recitation transmitted through recognised chains of oral instruction.

This oral dimension is essential for the present work. Many recitation errors cannot be fully understood from the written text alone. Two written sequences may differ in how they are realised acoustically depending on elongation, nasalisation, assimilation, concealment, or pausing. Automatic Tajweed assessment therefore requires a combination of textual reference, acoustic analysis, and rule-aware interpretation none of these three is sufficient on its own.

1.2.2 Qirā’āt, Riwayāt, and Recitation Variants

The Qur’an has been transmitted through recognised canonical readings known as *Qirā’āt*, each associated with a well-known imam of recitation and transmitted through specific narrations or *Riwayāt*. The most widely known examples include Ḥafṣ from ‘Āṣim and Warsh from Nāfi’, both of which differ in Hamzah treatment, vowel realisation, and Madd durations at specific positions. Academic studies on the history of Qur’anic transmission show that these traditions form a structured and documented science of variant reading [13]. Grodzki provides a detailed comparative discussion of printed Ḥafṣ and Warsh readings, documenting differences in vowel, Hamzah, and orthographic realisation [9].

For automatic assessment, this has a direct practical consequence: the expected pronunciation at any given position depends on which transmission the learner follows. A reference-aware system must therefore know not only the canonical text but also the applicable transmission-specific realisations. In this thesis, the focus is placed on Tajweed rules that are common and pedagogically central across traditions Madd, Ghunnah, Ikhfā’, Idghām, Iqlāb, and Qalqalah while the architecture remains compatible with transmission-specific reference data when needed.

1.2.3 Fundamental Tajweed Rules

Tajweed is the discipline that regulates the correct pronunciation and performance of Qur’anic recitation. It covers articulation points of letters, letter attributes, nasalisation, assimilation, concealment, elongation, pausing, and other rule-governed behaviours.

Table 1.1 presents the main Tajweed rule families, their representative examples, their acoustic relevance for assessment, and the category of acoustic analysis they require. This last column is of particular importance: it shows that Tajweed errors are not acoustically uniform, and that

different rule families require fundamentally different forms of detection.

Table 1.1: Common Tajweed rule families, their acoustic relevance, and the type of analysis they require.

Rule family	Examples	Why it matters for assessment	Analysis category
Makhārij al-Ḥurūf	Correct articulation points of Arabic letters	Detecting letter substitutions and articulation errors.	Content / articulatory
Sifāt al-Ḥurūf	Hams, Jahr, Shid-dah, Rikhāwah, Tafkhīm, Tarqīq	Detecting acoustic qualities such as emphasis, friction, and strength.	Local spectral
Nūn Sākinah and Tanwīn	Izhār, Idghām, Iqlāb, Ikhfāʾ	Context, nasalisation, and transitions between sounds.	Transition-oriented
Mīm Sākinah	Idghām Shafawī, Ikhfāʾ Shafawī, Izhār Shafawī	Lip-related assimilation and concealment.	Transition-oriented
Madd	Natural, connected, separated, and necessary elongation	Sustained duration of a vowel over a required interval.	Duration-oriented
Ghunnah	Nasalisation in specific contexts	Nasal resonance and its duration.	Duration-oriented
Qalqalah	Minor and major Qalqalah	Short localised release or echo-like acoustic event.	Burst-oriented
Waqf and Ibtidāʾ	Pausing and restarting	Phrase boundaries and acceptable stopping points.	Contextual / structural

The four analysis categories in the rightmost column duration, transition, burst, and contextual are not an arbitrary classification. They reflect the distinct types of acoustic evidence required to evaluate each rule family, and they recur as organising principles throughout this thesis.

1.2.4 Types and Severity of Recitation Errors

Errors in Qur’anic recitation differ in nature and severity. The classical pedagogical distinction is between *Lahn Jalī* and *Lahn Khafī*.

Lahn Jalī. *Lahn Jalī* refers to an obvious or major error, typically involving a deviation from the canonical text itself. It may involve replacing one letter with another, omitting part of a word, changing a vowel in a way that affects grammatical meaning, or otherwise altering the recited content. These errors are more severe because they affect *what* was recited, and they are therefore primarily a content-level problem.

Lahn Khafī. *Lahn Khafī* refers to a subtler error that concerns *how* the correct content was recited. Examples include shortening a Madd below its required duration, weakening a Ghunnah, failing to realise Ikhfā’, or omitting Qalqalah. These errors leave the word itself intact but violate the expected Tajweed realisation at that position.

This two-level distinction has a direct consequence for system design. A system that only detects content errors misses the entire *Lahn Khafī* class. A system that analyses rule execution without first verifying content may produce rule judgments on incorrectly recited segments. A complete assessment system must handle both levels and keep them logically ordered.

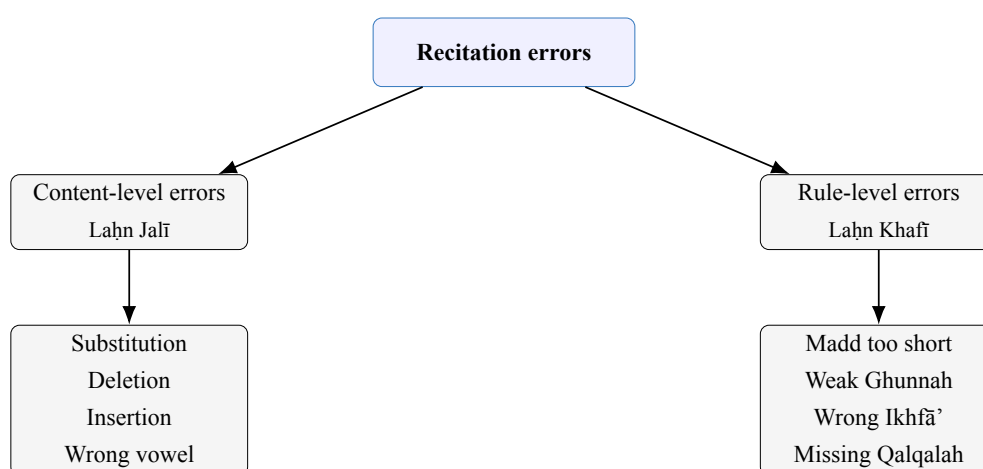


Figure 1.1: Taxonomy of recitation errors distinguishing content-level deviations (*Lahn Jalī*) from rule-level misapplications (*Lahn Khafī*).

1.2.5 Traditional Learning and Teacher Correction

Traditional Qur’anic learning relies on listening, repetition, memorisation, and correction by a qualified teacher. The learner listens to a model recitation, repeats it in small segments, receives correction, and revises the passage until the recitation is stable.

The diagnostic value of the teacher lies not only in detecting that an error exists, but in performing three distinct functions simultaneously: identifying *where* in the verse the error oc-

curred, determining *what type* of error it is whether it concerns the content, a duration rule, a transition rule, or another Tajweed category and explaining *how* it should be corrected. These three functions localisation, classification, and corrective guidance define the minimum specification of an automated assessment system that aims to replace or complement the teacher's role. A system that only signals the presence of an error without locating it or naming its type provides far less pedagogical value than a qualified human instructor.

1.2.6 The Coloured Mushaf as a Visual Learning Aid

The coloured Mushaf, or *Mushaf al-Tajweed al-Mulawwan*, is a widely used printed learning aid in which colours mark the positions where specific Tajweed rules should be applied. The colour does not alter the Qur'anic text; it encodes the rule attached to a letter or word and makes it immediately visible to the reader.

Colour conventions vary between publishers, but the general principle is consistent: one colour may indicate Madd, another Ghunnah or Idghām, another Qalqalah, and another letters that are written but not pronounced. This visual support is particularly valuable for beginners because many rules depend on context and are not derivable from the plain text alone. The coloured Mushaf therefore provides a rule-aware layer on top of the canonical text, guiding the learner from the written position to the expected recitation behaviour. Published editions such as the Dar al-Ma'rifah *Mushaf al-Tajweed al-Mulawwan* represent this pedagogical tradition [7].

1.3 Resources, Related Work, and Research Gap

1.3.1 Speech and Language Resources for Qur’anic Recitation

Developing an automatic Tajweed assessment system requires several distinct categories of resources. Audio corpora are needed to train or adapt acoustic models. Canonical text resources define the expected recitation. Rule annotations identify the positions where Tajweed rules apply. Error-labelled datasets are required to evaluate mispronunciation and rule-realisation detection.

Expert recitation audio. Recordings from expert reciters provide clean examples of correct pronunciation, rhythm, and recitation style, and support acoustic modelling or reference-based analysis. They are not, however, sufficient for error detection in isolation, since they contain correct recitation only and carry no information about how errors manifest.

Textual and linguistic resources. Textual resources such as the Qur’anic Arabic Corpus provide linguistic information including word-level and morphological structure. For Tajweed assessment specifically, the most important textual resource is not the plain canonical text but a rule-annotated reference that indicates where Tajweed rules apply and which rule governs each position. Such a resource is what allows the system to move from generic pronunciation scoring to rule-specific diagnosis.

Benchmarks and shared evaluation resources. A significant recent development is the emergence of standardised benchmarks for Qur’anic pronunciation assessment. El Kheir et al. introduced QuranMB.v1, the first unified benchmark for Qur’anic mispronunciation detection, together with self-supervised baseline models evaluated under shared conditions [8]. The IqraEval shared task provided a further evaluation framework for Arabic and Qur’anic phoneme-level pronunciation assessment; Ibrahim introduced Hafs2Vec, a system targeting this task at the phoneme level and reporting competitive performance under the shared evaluation protocol [11]. The Tarteel dataset, introduced by Khan et al., is a large crowd-sourced Qur’anic recitation corpus containing over 25,000 audio clips and 67 hours of audio contributed by more than 1,200 speakers, which has supported downstream research on Qur’anic speech technology [12]. These resources collectively indicate that the field is moving toward standardised, reproducible evaluation rather than isolated single-task experiments.

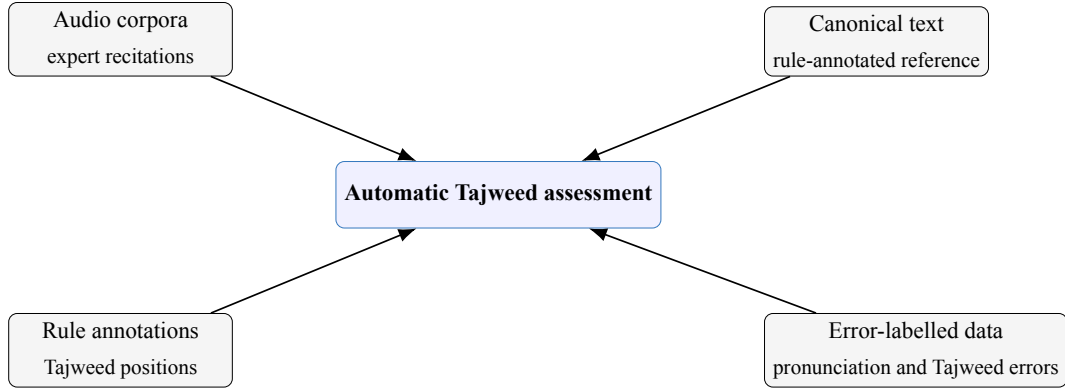


Figure 1.3: Main resource types required for automatic Tajweed assessment.

1.3.2 General Landscape of Related Work

Existing work relevant to automatic Qur’anic recitation assessment can be grouped into four broad directions. Some studies focus on Qur’anic speech recognition or verse identification. Others target the detection of specific Tajweed rules through acoustic classification. A third direction comes from general Arabic mispronunciation detection, where the task is to identify phoneme-level pronunciation errors in Arabic speech by non-native or learner speakers. More recent work explores self-supervised speech models, multimodal fusion architectures, and feedback generation.

Each direction addresses only part of the complete assessment problem. Speech recognition establishes what was recited. Tajweed classification evaluates rule-level correctness for specific rules. Arabic mispronunciation detection provides methods for phoneme-level error identification. Feedback-oriented systems show how detection output can be turned into learner guidance. A complete Tajweed assessment system must integrate all of these directions rather than rely on any single one.

1.3.3 Existing Approaches to Qur’anic Recitation Assessment

Feature-engineering and classical classification. The earliest line of Tajweed assessment research relied on hand-crafted acoustic features combined with classical machine-learning classifiers. Al-Ayyoub et al. represent the most ambitious system in this category: they target eight Tajweed rules across the entire Qur’an, formulating the problem as a 16-class task (eight rules, each labelled correct or incorrect) [2]. Their system combines MFCCs, wavelet packet decomposition, and HMM spectral peak features with a Convolutional Deep Belief Network (CDBN) for unsupervised representation learning, followed by an SVM classifier. On a dataset of 3,071 audio clips from ten reciters, the combined system achieves 97.8% classification accuracy. The main limitations are its reliance on a private dataset, a high computational cost of 15 seconds per clip, and the absence of any feedback mechanism: the system outputs a class label only.

MFCC-based sequential modelling. A second direction uses MFCC features with recurrent architectures designed to model the temporal structure of speech. This approach is particularly well suited to Tajweed because many rules are inherently temporal: Madd requires the system to track how long a vowel is sustained, and Ghunnah requires detection of nasal duration over several frames. Al Harere and Al Jallad trained separate LSTM models for three Tajweed rules – Madd, Ghunnah, and Ikhfā’ – on the public QDAT dataset of over 1,500 audio clips [3]. Using 40-coefficient MFCCs and a three-layer stacked LSTM, they achieve 96% accuracy on Madd and Ghunnah, and 95% on Ikhfā’, with F1 scores of 95–97% across the three rules. These results consistently outperform classical baselines (SVM, KNN, Random Forest) by margins of up to 12 percentage points on the same dataset. Ahmed et al. extend this direction to general Arabic mispronunciation detection, proposing a two-level LSTM framework that first detects the presence of a mispronunciation and then classifies its type [1]. Their MFCC-based BiLSTM achieves strong performance on Arabic learner speech and demonstrates that sequential modelling benefits from the bidirectional architecture when the judgment depends on both preceding and following phonetic context.

CNN-based local acoustic modelling. Convolutional architectures treat the acoustic representation as a spatial pattern-learning problem. They are effective at capturing compact local events and time-frequency patterns, which makes them naturally suited to rules whose acoustic evidence is concentrated in a narrow temporal window. Salim et al. apply a CNN to MFCC feature maps for Tajweed law recognition and conduct a systematic experiment on training set ratios, showing that model performance is sensitive to the proportion of labelled data used during training [14]. Their results confirm that CNNs are competitive for local classification tasks but less well suited to rules requiring long-range temporal dependencies unless combined with a sequential layer.

Multimodal and fusion-based approaches. A separate line of work explores combining complementary representations through multi-branch fusion architectures. Hassan et al. propose the Voice Shortcut Connection Fusion (VSCF) model, which runs MFCC features through a compact CNN and an audio representation through a ResNet50 encoder in parallel, then concatenates their outputs before a classification head [10]. Applied to Qur’anic reciter identification on a dataset of 7,144 audio clips from twelve reciters, the fused system achieves 97.68% accuracy, outperforming single-branch CNN baselines by more than one percentage point. Although the task is reciter classification rather than Tajweed error detection, the VSCF architecture is directly relevant because it demonstrates that combining local MFCC-based features with broader representation-level features improves acoustic discrimination.

Self-supervised and Transformer-based approaches. Recent Arabic mispronunciation detection has shifted toward self-supervised speech encoders and Transformer-based architectures.

Alrashoudi et al. fine-tune three large pre-trained models: wav2vec 2.0 XLS-R (300M parameters), HuBERT Large, and Whisper-small – on Arabic learner speech for phoneme-level mispronunciation detection [4]. The fine-tuned wav2vec 2.0 model achieves a phoneme error rate of 3.2%, outperforming human annotators on several error categories. Their work shows that self-supervised encoders pre-trained on large multilingual corpora transfer effectively to Arabic phoneme recognition without requiring a large Tajweed-specific labelled corpus. Çalik et al. independently confirm this direction, introducing an audio-oriented transformer framework for Arabic phoneme mispronunciation detection and demonstrating competitive results on Arabic speaker data [6]. Collectively, these results establish self-supervised encoders as the current state of the art for content recognition and alignment tasks in Arabic speech assessment.

Deployed Qur’anic learning applications. Beyond research systems, several deployed applications demonstrate the practical demand for automatic recitation support. Tarteel is the most widely used, offering verse verification, memorisation tracking, and immediate feedback on recitation content. The Tarteel dataset, introduced by Khan et al., contains more than 25,000 audio clips totalling 67 hours from over 1,200 contributors, and has supported downstream research [12]. Other applications reviewed by Birgün include TajweedMate, which offers colour-coded Tajweed detection, and Qur’an Tutor, which provides pronunciation correction [5]. Birgün’s review reveals a consistent pattern across deployed tools: they achieve reasonably strong recognition but provide limited or superficial feedback: they signal that an error occurred but do not explain its nature or how to correct it. No reviewed application produces rule-specific, position-aware diagnostic output.

Feedback-oriented pronunciation systems. The limitation identified in deployed applications, namely detection without explanation, has been recognised in the broader mispronunciation research community. Xie et al. present a cascaded system combining acoustic mispronunciation detection with an LLM-based feedback generator, and show that fine-tuning a language model on articulatory descriptions can produce corrective guidance that is preferred by listeners [15]. Their work is not specific to Arabic or Qur’anic recitation and remains primarily a proof of concept, but it demonstrates that the connection between acoustic error detection and linguistically grounded corrective feedback is technically feasible.

1.3.4 Evolution of Approaches

Figure 1.4 places the reviewed work in chronological perspective. The field has moved through three broad phases: a classical phase relying on hand-crafted features and conventional classifiers (2018–2020), a recurrent phase in which MFCC-based LSTM models became the dominant baseline for temporal rule analysis (2021–2023), and a current phase in which self-supervised encoders and Transformer-based architectures define the state of the art for content recognition

and alignment (2024–present).

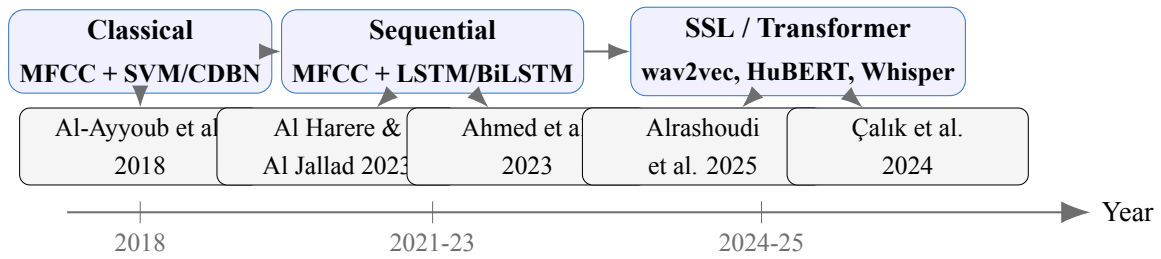


Figure 1.4: Evolution of modelling approaches in Qur’anic recitation assessment and Arabic mispronunciation detection.

1.3.5 Comparative Summary of Reviewed Systems

Table 1.2 synthesises the main characteristics of the systems reviewed above across a set of criteria directly relevant to the design of a complete automatic Tajweed assessment system.

Table 1.2: Comparative summary of reviewed systems for Qur’anic recitation assessment and Arabic mispronunciation detection.

Work	Year	Approach	Task / Rules	Dataset	Feedback	Key limitation
Al-Ayyoub et al.	2018	MFCC + CDBN + SVM	8 rules, 16 classes	In-house (3,071)	None	Private data; no feedback; out-dated SSL
Al Harere & Al Jallad	2023	MFCC + LSTM	3 rules (binary)	QDAT (1,500+)	None	3 rules only; single verse
Ahmed et al.	2023	MFCC + BiLSTM	Arabic MDD (phoneme)	In-house	None	Not Tajweed-specific
Salim et al.	2025	MFCC + CNN	Tajweed law recognition	In-house	None	Local patterns only; limited rule set
Hassan et al.	2025	MFCC + CNN / ResNet50 fusion	Reciter classification	7,144 clips	None	Task mismatch (not error detection)
Alrashoudi et al.	2025	wav2vec / HuBERT / Whisper fine-tuning	Arabic MDD (phoneme)	L2-KSU	None	Not Tajweed-specific; no rule routing
Çalik et al.	2024	Transformer (audio-oriented)	Arabic phoneme MDD	Arabic speaker data	None	Not Tajweed-specific

1.3.6 Comparison of Model Architecture Options

Table 1.3 summarises the main model architectures found in the literature, their strengths and weaknesses, and the evidence supporting each.

Table 1.3: Comparison of model architectures discussed in the reviewed literature.

Model	Strengths	Weaknesses	Evidence from the literature
CNN 1D/2D	Efficient for local acoustic and time-frequency patterns; suitable for short events.	Limited temporal range unless combined with a sequential layer.	MFCC–CNN used for Tajweed law recognition; performance degrades on acoustically similar classes [14].
LSTM / BiLSTM	Designed for sequence data, variable-length input, and temporal dependencies; well suited to duration and transition rules.	Slower to train; may struggle with very long sequences.	MFCC–LSTM achieves 95–97% F1 on three Tajweed rules [3]; BiLSTM benefits from bidirectional context for Arabic MDD [1].
SSL encoder (wav2vec, HuBERT, Whisper)	Rich contextual representations; strong phoneme recognition; robust to speaker and recording variation; effective under limited labelled data.	Computationally expensive; requires large pre-trained models.	Fine-tuned wav2vec achieves 3.2% PER, outperforming human raters on Arabic [4]; Transformer models show competitive results on Arabic phoneme MDD [6].
Multimodal fusion	Combines local and global representations; improves discrimination when evidence is complementary across feature scales.	Architecture complexity; requires parallel branches.	VSCF fusion of MFCC and ResNet50 outperforms single-branch CNN by 1+ percentage points on Qur’anic audio [10].

The pattern across Table 1.3 is consistent: no single architecture is universally optimal. CNNs handle local patterns well, recurrent models capture temporal structure, and SSL encoders excel at content recognition and alignment. This supports a modular design strategy in which each rule category is analysed by the architecture best matched to its dominant acoustic evidence.

1.3.7 Research Gap and Positioning of the Present Work

The state of the art, as summarised in Table 1.2, reveals a consistent pattern. Existing systems make progress on individual sub-problems: feature extraction, rule classification, phoneme-level detection. But none integrates all of these into a unified assessment pipeline that also

connects detection to localisation, rule identification, and corrective feedback.

Three specific gaps are identifiable.

Coverage and granularity. Most Tajweed-specific systems cover only a small number of rules and produce binary correct/incorrect outputs. A learner who receives a binary verdict knows that something was wrong but cannot act on that information without knowing where or why.

Separation of content and rule errors. No reviewed system explicitly separates content-level errors (Laḥn Jalī) from rule-level errors (Laḥn Khafī) in its diagnostic output. This distinction is pedagogically essential: the corrective guidance appropriate for a content substitution is entirely different from that appropriate for a shortened Madd.

Rule-aware feedback. No reviewed system links its detection output to a rule-annotated canonical reference to produce named, position-aware diagnostic feedback. Systems either stop at detection or rely on a general feedback generator that operates without specific knowledge of which Tajweed rule was violated at which canonical position.

Two broader perspectives complement this picture without belonging to the same comparison class. Birgün’s qualitative review of deployed applications confirms that the detection-without-explanation limitation is not specific to any single system but is a field-wide pattern across all reviewed tools [5]. Xie et al.’s work on LLM-based feedback generation demonstrates that connecting acoustic error detection to linguistically grounded corrective output is technically feasible, even though their system targets English and Chinese speech and remains a proof of concept rather than a deployed Tajweed tool [15]. Both works reinforce the same conclusion: the gap is real, it is recognised across the field, and closing it requires a system that goes beyond detection.

This thesis addresses all three gaps. It treats automatic Tajweed assessment as a modular and reference-aware diagnostic task rather than a global classification problem. The architecture separates content verification from rule analysis, routes each rule instance to the appropriate specialist module, and uses an annotated Qur’anic reference to convert detection outputs into named, position-specific corrective feedback.

1.3.8 Chapter Summary

This chapter reviewed the linguistic, pedagogical, and technical background necessary for understanding automatic Tajweed assessment. It introduced Qur’anic recitation and the system of canonical transmission traditions, presented the main Tajweed rule families and their acoustic categories, described the two levels of recitation error (Laḥn Jalī and Laḥn Khafī), and discussed the pedagogical role of traditional instruction and the coloured Mushaf.

The review of related work showed that the field has progressed through three modelling phases: classical feature engineering, recurrent sequence modelling, and self-supervised representation learning. No existing system integrates content verification, rule-aware routing,

structured diagnosis, and corrective feedback into a single pipeline. This gap directly motivates the modular, reference-aware architecture developed in the following chapter.

s

Bibliography

- [1] Abdelfatah Ahmed, Mohamed Bader, Ismail Shahin, Ali Bou Nassif, Naoufel Werghi, and Mohammad Basel. Arabic mispronunciation recognition system using LSTM network. *Information*, 14(7):413, 2023. doi: 10.3390/info14070413.
- [2] Mahmoud Al-Ayyoub, Nour Alhuda Damer, and Ismail Hmeidi. Using deep learning for automatically determining correct application of basic Quranic recitation rules. *International Arab Journal of Information Technology*, 15(3A):620–625, 2018.
- [3] Ahmad Al Harere and Khlood Al Jallad. Mispronunciation detection of basic Quranic recitation rules using deep learning. *arXiv preprint arXiv:2305.06429*, 2023. URL <https://arxiv.org/abs/2305.06429>.
- [4] Norah Alrashoudi, Hend Al-Khalifa, and Yousef Alotaibi. Improving mispronunciation detection and diagnosis for non-native learners of the arabic language. *Discover Computing*, 28(1):1, 2025. doi: 10.1007/s10791-024-09489-8.
- [5] Sener Birgun. A review of automated Quranic recitation assessment systems. *Journal of Islamic Studies and Technology*, 2026.
- [6] Şükrü Selim Çalık, Ayhan Küçükmanisa, and Zeynep Hilal Kilimci. A novel framework for mispronunciation detection of arabic phonemes using audio-oriented transformer models. *Applied Acoustics*, 215:109711, 2024. doi: 10.1016/j.apacoust.2023.109711.
- [7] Dar al-Ma’rifah. *Mushaf al-Tajweed al-Mulawwan*. Dar al-Ma’rifah, Beirut, 2001.
- [8] Yassine El Kheir, Omnia Ibrahim, Amit Meghanani, Nada Almarwani, Hawau Olamide Toyin, Sadeen Alharbi, Modar Alfadly, Lamya Alkanhal, Ibrahim Selim, Shehab Elbatal, Salima Mdhaftar, Thomas Hain, Yasser Hifny, Mostafa Shahin, and Ahmed Ali. Towards a unified benchmark for arabic pronunciation assessment: Qur’anic recitation as case study, 2025. Accepted at Interspeech 2025 and ArabicNLP Shared Task 2025.
- [9] Marcin Grodzki. Between the lines of the hafs and warsh readings of the qur’an: The qur’an as a literary testimony to its meanings. *Rocznik Orientalistyczny / Yearbook of Oriental Studies*, 75(1):31–41, 2022. doi: 10.24425/ro.2022.141413.

- [10] Esraa Hassan, Abeer Saber, Omar Alqahtani, Nora El-Rashidy, and Samar Elbedwehy. An innovative approach to advanced voice classification of sacred Quranic recitations through multimodal fusion. *Egyptian Informatics Journal*, 30:100640, 2025. doi: 10.1016/j.eij.2025.100640.
- [11] Ahmed Ibrahim. Hafs2Vec: A system for the IqraEval Arabic and Qur’anic phoneme-level pronunciation assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 453–456, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.arabicnlp-sharedtasks.62.
- [12] Hamzah I. Khan, Abubakar Abid, Mohamed Medhat Moussa, and Anas Abou-Allaban. The tarteel dataset: Crowd-sourced and labeled quranic recitation. OpenReview, 2021. URL <https://openreview.net/forum?id=TAdzPkgnnV8>. Submitted to NeurIPS 2021 Datasets and Benchmarks Track.
- [13] Shady Hekmat Nasser. *The Transmission of the Variant Readings of the Qur’an: The Problem of Tawatur and the Emergence of Shawadhdh*, volume 9 of *Texts and Studies on the Qur’an*. Brill, Leiden, 2012. ISBN 9789004241794.
- [14] Mas Muhammad Aqil Salim, Ani Dijah Rahajoe, and Anggraini Puspita Sari. Dataset ratio experiment for Tajweed law recognition using MFCC and CNN features. *JEEMECs*, 8 (2):91–97, 2025. doi: 10.26905/jeemecs.v8i2.14789.
- [15] Yanlu Xie, Huihang Zhong, Xuhui Lan, and Wenwei Dong. Mispronunciation detection and diagnosis based on large language models. *Computer Speech & Language*, 99:101942, 2026. doi: 10.1016/j.csl.2026.101942.